

MITHILESH KUMAR U S

+91-89039-89225 | mithilesh02905@gmail.com | linkedin.com/in/mithileshkumarus | github.com/mithilesh-kumar-us | Portfolio

EDUCATION

Amrita Vishwa Vidyapeetham

Chennai, India

Bachelor of Technology (B.Tech.) in Computer and Communication Engineering – **CGPA: 8.91/10.0** August 2023 – May 2027

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems, Computer Networks, Machine Learning, Data Science, Deep Learning, Distributed Systems

Perks Matriculation School

Coimbatore, India

Higher Secondary Education (HSE), Computer Science – **Percentage: 92.8%**

June 2021 – May 2023

EXPERIENCE

SAP Labs India – Software Engineering Intern (iXp)

Bengaluru, India | May 2026 – July 2026

- Reduced manual documentation lookup time by **40%** by building a **knowledge base platform** with an integrated **MCP server**, enabling engineers to query cross-team workflows, database schemas, and API details conversationally via **LLM-based tools** like **Claude Code**.
- Optimized manual API testing time by **60–70%** across **15+ endpoints** by engineering an **AI workflow** that ingests endpoint specifications and documentation, auto-generates test scenarios, and executes them via **MCP tools** against a live environment, delivering validated end-to-end API responses.

PingUs – Freelancer

Chennai, India | October 2025 – February 2026

- Replaced a cloud kitchen's manual, phone-based ordering process with a full-stack web platform handling **5–10 daily orders**, eliminating manual order-entry errors through structured menu browsing, cart, and checkout flows.
- Built an **admin dashboard** for real-time menu and order management, backed by a React + Tailwind CSS frontend and a Node.js (Express) + MongoDB backend.

EasyGold Pvt. Ltd. – Web Developer Intern

Remote | April 2025 – June 2025

- Built a real-time **e-gold product description page** for a beta launch, integrating a **live pricing API** refreshing **every few minutes** and a secure checkout flow across **mobile and desktop** – end-to-end in **React.js** and **Node.js**.

PROJECTS

PRISM | Python, FastAPI, Redis, PostgreSQL, Next.js, TypeScript, Docker, CI/CD Pipeline

GitHub June 2026

- Integrated a queue-driven AI code review pipeline supporting **11 languages** via **Tree-sitter AST parsing** and a **resolved call graph**, running 7 static analyzers in **isolated Docker sandboxes** for zero-trust PR analysis.
- Engineered a bounded tool-calling agent across LLM providers with an independent judge-validation pass, and shipped a Next.js dashboard with **GitHub OAuth** on a **FastAPI + PostgreSQL** backend.

SyncSpace | Node.js, Express, MongoDB, Redis, BullMQ, WebSocket, Yjs, Next.js, TypeScript

GitHub January 2026

- Architected a sync engine around **Yjs CRDTs** with a custom **WebSocket** layer, scaled horizontally via a **Redis pub/sub** relay, with permissions enforced at the raw **WebSocket message** level.
- Delivered a state-reconciliation bridge extending that sync core to a collaborative Excalidraw whiteboard, resolving concurrent edits via element-level tombstoning, a **50ms-throttled** shape sync, and live multi-cursor presence tracking.

TECHNICAL SKILLS

Languages: C++, Python, Java, SQL

Backend: Node.js, Express.js, FastAPI, REST APIs, Microservices, Distributed Systems, Linux

Frameworks: React.js, Next.js, Tailwind CSS

Data / AI: PyTorch, LangChain, LangGraph, TensorFlow, Transformers, pgvector, LLMs, RAG, MCP

Databases: PostgreSQL, MySQL, MongoDB, Redis

Cloud / DevOps: AWS, Docker, CI/CD Pipelines (GitHub Actions)

ACHIEVEMENTS & CERTIFICATIONS

- **Won 6+ hackathon competitions** across software engineering and AI-focused challenges.
- **Winner – TechBiz'25**, National Level Hackathon – built an **AI proctoring system** with real-time face and gaze tracking, competing against **60+ teams**.
- **Finalist – IndustriAI Hackathon**, IIT Madras – selected among **300+ teams**; deployed an **AI-driven detection system**.
- **Certifications:** AWS Cloud Foundations, Cisco Networking Basics, Generative AI (DeepLearning.AI), Problem Solving (HackerRank).